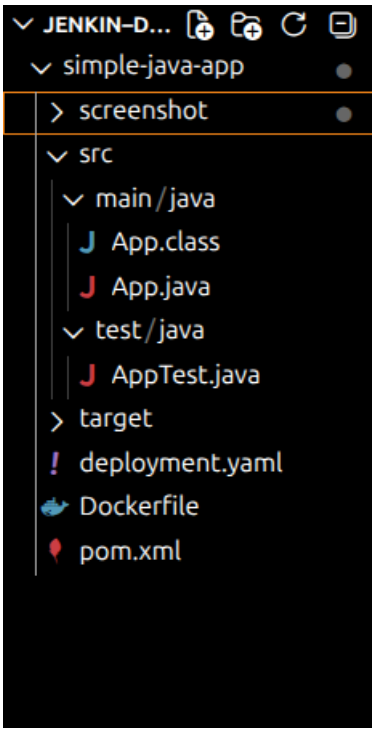
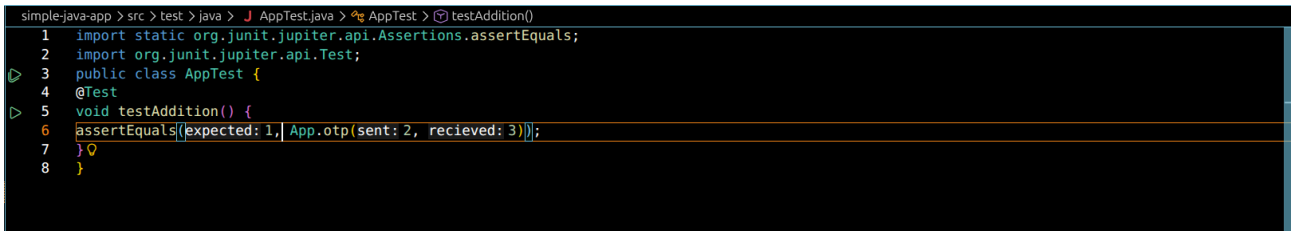
Kanish kumar S
23MIS0141
Agile Development and Devops Lab Fat

Create project folder



App.java



AppTest.java

```
simple-java-app > src > main > java > App.java > App > main(String[])
public class App {
    public static int otp(int sent, int recieved) {
        if(sent==recieved)
        {
            return 0;
        }
        else
        {
            return 1;
        }

    }
    Run | Debug
    public static void main(String[] args) {
        System.out.println(otp(sent: 2,recieved: 3));
    }
}
```

Mvn clean test

```
~/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app$mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ test-java-app ---
[INFO] Deleting /home/kanish-kumar/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ test-java-app ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /home/kanish-kumar/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug target 17] to target/classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ test-java-app ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /home/kanish-kumar/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug target 17] to target/test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- maven-surefire-plugin:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.069 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.493 s
[INFO] Finished at: 2026-04-13T10:06:54+05:30
[INFO] ------------------------------------------------------------------------
```

```
~/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app$mvn clean package
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ test-java-app ---
[INFO] Deleting /home/kanish-kumar/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ test-java-app ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /home/kanish-kumar/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug target 17] to target/classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ test-java-app ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /home/kanish-kumar/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug target 17] to target/test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- maven-surefire-plugin:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.066 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.4.1:jar (default-jar) @ test-java-app ---
[INFO] Building jar: /home/kanish-kumar/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app/target/test-java-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.847 s
```

push to github

```
~/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app$git push origin master
~/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app$git push origin master
Everything up-to-date
~/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app$git add .
~/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app$git commit -m "p"
[master 7fdbe57] p
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 screenshot/Pasted image (12).png
~/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app$git push origin master
fatal: unable to access 'https://github.com/Kanish-Kumar-S/agile-2/': Could not resolve host: github.com
~/Public/agile-prac/Jenkin-Docker-Kubernetes-Junit/simple-java-app$git push origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 46.14 KiB | 7.69 MiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0
```

pipeline

```
Started by user Kanish Kumar S 23MIS0141
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/agile-proj
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Repository)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/agile-proj/.git # timeout=10
Fetching changes from the remote Git repository
 > git config remote.origin.url https://github.com/Kanish-Kumar-S/jenkin-docker-kuber # timeout=10
Fetching upstream changes from https://github.com/Kanish-Kumar-S/jenkin-docker-kuber
 > git --version # timeout=10
 > git --version # 'git version 2.43.0'
 > git fetch --tags --force --progress -- https://github.com/Kanish-Kumar-S/jenkin-docker-kuber +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision 8ba070280516c6f43b222c2c7c535e1dc1531c2e (refs/remotes/origin/master)
 > git config core.sparsecheckout # timeout=10
 > git checkout -f 8ba070280516c6f43b222c2c7c535e1dc1531c2e # timeout=10
 > git branch -a -v --no-abbrev # timeout=10
 > git branch -D master # timeout=10
 > git checkout -b master 8ba070280516c6f43b222c2c7c535e1dc1531c2e # timeout=10
Commit message: "4201"
 > git rev-list --no-walk 8ba070280516c6f43b222c2c7c535e1dc1531c2e # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build with Maven)
[Pipeline] sh
+ mvn clean package -DskipTests
```

create jenkin pipeline

Enter an item name

pipeline1

Select an item type

Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.

Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.

Duplicate an existing item
Create a new item by copying the configuration of an existing one.

OK

```
pipeline {
    agent any

    environment {
        DOCKER_IMAGE = "kanishkumar2005/java-app"
        DOCKER_TAG = "latest"
    }

    stages {

        stage('Clone Repository') {
            steps {
                git branch: 'master', url: 'https://github.com/Kanish-Kumar-S/jenkin-docker-kuber'
            }
        }

        stage('Build with Maven') {
            steps {
                sh 'mvn clean package -DskipTests'
            }
        }

        stage('Build Docker Image') {
            steps {
                sh 'docker build -t $DOCKER_IMAGE:$DOCKER_TAG .'
            }
        }

        stage('Login to DockerHub') {
            steps {
                withCredentials([usernamePassword(
                    credentialsId: 'dockerhub-creds',
                    usernameVariable: 'DOCKER_USER',
                    passwordVariable: 'DOCKER_PASS'
                )]) {
                    sh 'echo $DOCKER_PASS | docker login -u $DOCKER_USER --password-stdin'
                }
            }
        }

        stage('Push to DockerHub') {
            steps {
                sh 'docker push $DOCKER_IMAGE:$DOCKER_TAG'
            }
        }

        stage('Deploy to Kubernetes') {
            steps {
                sh 'kubectl apply -f deployment.yaml'
            }
        }
    }

    post {
        success {
            echo '✅ Pipeline executed successfully!'
        }
        failure {
            echo '❌ Pipeline failed. Check logs.'
        }
    }
}
```

```
a7428f4f5c1d: Waiting
a7428f4f5c1d: Waiting
a7428f4f5c1d: Waiting
a7428f4f5c1d: Waiting
a7428f4f5c1d: Layer already exists
409b30d41a0c: Pushed
latest: digest: sha256:270bf986a954a3a2337a08afe8ac3b2bb3f0b0776164f3c4bc896a80a5de1120 size: 856
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] sh
+ kubectl apply -f deployment.yaml
deployment.apps/java-app unchanged
service/java-app-service unchanged
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
✅ Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.558

registry.k8s.io/kube-proxy	v1.34.1	913cc83ca0b5	7 months ago	101.65 MB	
registry.k8s.io/etcd	3.6.4-0	e36c08168342	9 months ago	272.54 MB	
registry.k8s.io/pause	3.10.1	278fb9dbcca9	10 months ag	1.05 MB	
registry.k8s.io/coredns/coredns	v1.12.1	e8c262566636	1 year ago	100.71 MB	
registry.k8s.io/pause	3.10	ee6521f290b2	2 years ago	1.05 MB	
kanishkumar2005/java-app	latest	270bf986a954	1 hour ago	635.94 MB	
<none>	<none>	b258f3743d1d	12 hours ago	635.94 MB	

Showing 15 items